

Classical Mechanics Equation Sheet 2

Chapter 8 Two-Body Central-Force

Problems

- CM and Relative Coordinates

- ▶ $M = m_1 + m_2$
- ▶ $\mu = \frac{m_1 m_2}{M}$
- ▶ $\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{M}$
- ▶ $\vec{r} = \vec{r}_1 - \vec{r}_2$
- ▶ $\vec{r}_1 = \vec{R} + \frac{m_2}{M} \vec{r}$
- ▶ $\vec{r}_2 = \vec{R} - \frac{m_1}{M} \vec{r}$
- ▶ $T = \frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} \mu \dot{\vec{r}}^2$

- Kepler Orbits

- ▶ General
 - $r(\phi) = \frac{c}{1 + \epsilon \cos(\phi)}$
 - $c = \frac{l^2}{\gamma \mu}$
 - $E = \frac{\gamma^2 \mu}{2l^2} (\epsilon^2 - 1)$
 - $\gamma = GMm$ for gravitational force
 - $F(r) = -\frac{\gamma}{r^2}$
 - $U(r) = -\frac{\gamma}{r}$
- ▶ Bounded Orbits ($0 \leq \epsilon < 1$)
 - $\frac{(x+d)^2}{a^2} + \frac{y^2}{b^2} = 1$
 - $a = \frac{c}{1-\epsilon^2}$
 - $b = \frac{c}{\sqrt{1-\epsilon^2}}$
 - $d = a\epsilon$
 - $\tau^2 = \frac{4\pi^2 \mu}{\gamma} a^3$
- ▶ Unbounded Orbits ($\epsilon \geq 1$)
 - $y^2 = c^2 - 2cx$ when $\epsilon = 1$
 - $\frac{(x-\delta)^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ when $\epsilon > 1$
 - $\alpha = \frac{c}{\epsilon^2 - 1}$
 - $\beta = \frac{c}{\sqrt{\epsilon^2 - 1}}$
 - $\delta = \alpha\epsilon$

Chapter 9 Mechanics in Noninertial

Frames

- Inertial Force

- ▶ $\vec{F}_{\text{inertial}} = -m\vec{A}$

- Tides

- ▶ $\vec{F}_{\text{tid}} = -GM_m m \left(\frac{\hat{d}}{d^2} - \frac{\hat{d}_0}{d_0^2} \right)$
- ▶ $U_{\text{tid}} = -GM_m m \left(\frac{1}{d} + \frac{x}{d_0^2} \right)$

- Angular Velocity Vector

- ▶ $\vec{\omega} = \omega \hat{u}$
- ▶ $\vec{v} = \vec{\omega} \times \vec{r}$
- ▶ $\frac{d\hat{e}}{dt} = \vec{\omega} \times \hat{e}$
- ▶ $\vec{\omega}_{31} = \vec{\omega}_{32} + \vec{\omega}_{21}$

- Time Derivatives in a Rotating Frame

- ▶ $\left(\frac{d\vec{Q}}{dt} \right)_{S_0} = \left(\frac{d\vec{Q}}{dt} \right)_S + \vec{\Omega} \times \vec{Q}$

- Newton's Second Law in a Rotating Frame

- ▶ $m\ddot{\vec{r}} = \vec{F} + \vec{F}_{\text{cor}} + \vec{F}_{\text{cf}}$
- ▶ $\vec{F}_{\text{cor}} = 2m\dot{\vec{r}} \times \vec{\Omega}$
- ▶ $\vec{F}_{\text{cf}} = m(\vec{\Omega} \times \vec{r}) \times \vec{\Omega} = m\Omega^2 \rho \hat{\rho}$

- Free-Fall Acceleration

- ▶ $\vec{g} = \vec{g}_0 + \Omega^2 R \sin(\theta) \hat{\rho}$

Chapter 10 Rotational Motion of Rigid Bodies

- Properties of Center of Mass

- ▶ $\vec{F}^{\text{ext}} = M\ddot{\vec{R}}$
- ▶ $\vec{r}'_{\alpha} = \vec{r}_{\alpha} - \vec{R}$
- ▶ $\vec{L} = \vec{R} \times \vec{P} + \sum \vec{r}'_{\alpha} \times m_{\alpha} \dot{\vec{r}}'_{\alpha}$
- ▶ $\vec{L} = \vec{L}_{\text{orb}} + \vec{L}_{\text{spin}}$
- ▶ $\dot{\vec{L}}_{\text{orb}} = \vec{R} \times \vec{F}^{\text{ext}}$
- ▶ $\dot{\vec{L}}_{\text{spin}} = \vec{\Gamma}^{\text{ext}}(\text{about CM})$
- ▶ $T = \frac{1}{2} M \dot{\vec{R}}^2 + \frac{1}{2} \sum m_{\alpha} \dot{\vec{r}}'_{\alpha}{}^2$

- Rotation About Any Axis

- ▶ $\vec{L} = \mathbb{I} \vec{\omega}$
- ▶ $\mathbb{I} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix}$
- ▶ $\mathbb{I} = \tilde{\mathbb{I}}$
- ▶ $I_{xx} = \sum m_{\alpha} (y_{\alpha}^2 + z_{\alpha}^2)$
- ▶ $I_{xy} = -\sum m_{\alpha} x_{\alpha} y_{\alpha}$
- ▶ $T = \frac{1}{2} \vec{\omega} \cdot \vec{L}$

- Principal Axes of Inertia

- ▶ $\vec{L} = \lambda \vec{\omega}$

- $(\mathbb{I} - \lambda \mathbb{1})\vec{\omega} = 0$
 - $\det(\mathbb{I} - \lambda \mathbb{1}) = 0$
 - $\mathbb{I}' = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$
- Euler's Equations
 - $\dot{\vec{L}} + \vec{\omega} \times \vec{L} = \vec{\Gamma}$ (body frame)
 - $\lambda_1 \dot{\omega}_1 - (\lambda_2 - \lambda_3)\omega_2\omega_3 = \Gamma_1$
 - $\lambda_2 \dot{\omega}_2 - (\lambda_3 - \lambda_1)\omega_3\omega_1 = \Gamma_2$
 - $\lambda_3 \dot{\omega}_3 - (\lambda_1 - \lambda_2)\omega_1\omega_2 = \Gamma_3$
- Generalized Parallel Axis Theorem
 - Pivot point displaced by $\Delta\vec{r} = (\Delta x, \Delta y, \Delta z)$ from the CM
 - $I_{xx} = I_{xx}^{\text{cm}} + M(\Delta y^2 + \Delta z^2)$
 - $I_{yz} = I_{yz}^{\text{cm}} - M\Delta y\Delta z$
- Precession of a Top due to a Weak Torque
 - $\lambda_1 = \lambda_2 \neq \lambda_3$
 - $\dot{\hat{e}}_3 = \vec{\Omega} \times \hat{e}_3$
 - $\vec{\Omega} = \frac{MgR}{\lambda_3\omega} \hat{z}$
- Free Precession of a Top
 - $\lambda_1 = \lambda_2 \neq \lambda_3$
 - $\Omega_b = \frac{\lambda_1 - \lambda_3}{\lambda_1} \omega_3$
 - $\Omega_s = \frac{L}{\lambda_1}$
- Generalized Momentum or Conjugate Momentum
 - $p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$
- If $\frac{\partial \mathcal{H}}{\partial t} = 0$, then $\mathcal{H} = \text{const.}$
- Hamilton's Equations
 - $\dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i}$
 - $\dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i}$
- Liouville's Theorem
 - $\vec{z} = (\vec{q}, \vec{p}) = (q_1, \dots, q_n, p_1, \dots, p_n)$
 - $\vec{v} = \dot{\vec{z}} = (\dot{\vec{q}}, \dot{\vec{p}})$
 - $\vec{\nabla} \cdot \vec{v} = 0$

Chapter 11 Coupled Oscillators and

Normal Modes

- Equations of Motion
 - $\mathbb{M}\ddot{\vec{q}} = -\mathbb{K}\vec{q}$
- Normal Modes
 - $\vec{q}(t) = \text{Re}[\vec{z}(t)]$
 - $\vec{z}(t) = \vec{a}e^{i\omega t}$
 - $(\mathbb{K} - \omega^2\mathbb{M})\vec{a} = 0$
 - $\det(\mathbb{K} - \omega^2\mathbb{M}) = 0$

Chapter 13 Hamiltonian Mechanics

- Lagrangian
 - $\mathcal{L} = T - U$
- Hamiltonian
 - $\mathcal{H} = \sum p_i \dot{q}_i - \mathcal{L}$
 - $\mathcal{H} = T + U$ if the generalized coordinates are “natural”